

NAAC Accreditation RAG Agent

Problem Statement

An AI agent designed to assist faculty with NAAC accreditation processes, powered by Retrieval-Augmented Generation (RAG). The agent retrieves institutional data, past reports, and NAAC guidelines to provide precise, context-aware support. Faculty can interact with the agent to generate SSR drafts, criteria-wise inputs, and best practice documentation. The RAG approach ensures answers are grounded in trusted sources and tailored to the institution's context, reducing manual effort and improving accuracy in accreditation readiness.

Why I Chose This Problem Statement

I picked this problem statement mainly because it let me apply something I was already learning on my own — I had recently started exploring RAG systems and MongoDB alongside a computer vision project I was building, so this felt like a natural next step rather than starting a completely new skill from scratch under internship deadline pressure.

The other reason was practicality. My own college is an autonomous institution, which meant I had direct access to the actual NAAC manual and real accreditation criteria instead of having to work with a synthetic or guessed dataset. A lot of the other problem statements in the list (health symptom checkers, nutrition agents, farming advice agents) would have needed me to fabricate realistic data, which felt like it would weaken the credibility of the final demo. With NAAC, I could build something grounded in a real, publicly available document and test it with genuine accreditation questions.

I also wanted a project that went beyond just wrapping an LLM in a chat interface. NAAC accreditation is a compliance-heavy process where a wrong or hallucinated answer could genuinely mislead a faculty member preparing their SSR, so this problem statement gave me a reason to actually think about grounding, retrieval accuracy, and citation — not just generate plausible-sounding text.

Proposed Solution

The solution is a Retrieval-Augmented Generation agent built on IBM watsonx.ai and the IBM Granite model, using the NAAC Revised Manual for Autonomous Colleges as its sole knowledge source. Instead of letting the model answer from its own general training, the manual itself is parsed and broken down into chunks aligned with its actual structure — Criterion, Key Indicator, and individual Metric — rather than being split by arbitrary page breaks. This keeps each retrievable unit meaningful, since a Key Indicator's description and its associated metrics stay together instead of being separated across chunk boundaries.

These chunks are converted into vector embeddings and stored for semantic search. When a faculty member asks a question in plain language — for example, 'What is required under

Key Indicator 3.4?' — the system retrieves the most relevant chunks based on meaning rather than exact keyword matching. The retrieved content is then passed to IBM Granite, which generates a response grounded strictly in that content, explicitly citing the Criterion, Key Indicator, and Metric number it came from. If nothing relevant is found in the manual, the agent says so directly instead of generating a confident but incorrect answer.

The system is accessible through a simple web interface with a text input for questions and a display area for the cited answer, and the entire pipeline runs on IBM Cloud Lite services, keeping it within the IBM ecosystem as required by the problem statement's technology constraint.

Expected Outcome

The end result is a tool that reduces the time faculty spend manually searching through a 153-page manual for specific requirements, while staying accountable — every answer can be checked against the exact section of the manual it claims to be based on. This makes it practically usable for real accreditation preparation, not just a demonstration of RAG technology.